

Author Response for Submission
“PushAround: Collaborative Path Clearing via
Physics-Informed Hybrid Search”

Author Names Omitted for Anonymous Review.

To Associate Editor

Q: *The reviewers acknowledge the contribution of the paper, however, they have some concerns that need to be addressed.*

A: We sincerely thank the Associate Editor for the careful handling of our submission and for summarizing the reviewers’ constructive comments. Both reviewers explicitly support the contribution and frame their comments as clarifications rather than technical objections, so the revision keeps the scope of the method unchanged. In response, we have substantially revised the manuscript to clarify the methodological details, implementation choices, and experimental limitations identified by the reviewers: the two formal weaknesses of Reviewer 3 are closed with dedicated experiments, and all follow-up questions and clarifications of both reviewers are answered explicitly in the manuscript. The principal revisions are summarized below.

Table 1: New and revised experiments at a glance. “Ours” is PushAround.

Study	Setting (workspace, M , N ; trials)	Ours	Strongest baseline	Location
Main comparison	8×5 m, $M=10$, $N=2$; 40 trials	97.5 %	SL-Push (sim) 75.0 %	Table II
Ablations	same scene; 40 trials	82.5–97.5 %	—	Table II
Obstacle density	8×8 m, $M=15$ –45, $N=2$; 10 seeds per density	100 % (all M)	collapses to 0 % at $M=45$	Table III, Fig. 10
Physics mismatch	8×8 m, $M=30$, $N=2$; $\pm 20\%$ / $\pm 40\%$; 10 (ours) / 5 (base) seeds	97.5 % at $\pm 40\%$	DFS-WCCG 57.5 %, SL-Push (sim) 32.5 %	Table III
Robot-team scaling	8×8 m, $M=30$, $N=2$ –4; 10 seeds per cell	100 % (30/30 trials)	DFS-WCCG 90–100 %, SL-Push (sim) 80 % (1.6–1.8 $\times$ slower execution)	Table III
WCCG validation	300 random scenes, incl. non-convex	300/300, 0 FP/FN	—	Table I

The principal revisions are:

- Planning uses the nominal model; only the execution simulator is perturbed, one parameter family at a time (obstacle mass, lateral friction, spinning friction, robot force limit) by factors $\{0.6, 0.8, 1.2, 1.4\}$, i.e. $\pm 20\%$ and $\pm 40\%$. PushAround succeeds in **166/170** trials overall (**97.5 %** at $\pm 40\%$ and **97.5 %** at $\pm 20\%$), whereas under the same $\pm 40\%$ perturbations SL-Push (sim) drops to **32.5 %** (13/40, including two conditions at 0%) and DFS-WCCG to **57.5 %** (23/40) with roughly an order of magnitude larger planning time (Sec. IV-A6, Table III, Fig. 1).
- The density sweep now uses **10 independent seed sequences** per density (previously 5). PushAround keeps **100 %** success for every $M \in [15, 45]$ with mean planning time growing from 2.5 to 53.4 s, while all four baselines degrade sharply (Fig. 2); the reported statistics include dispersion and Wilson confidence intervals (Sec. IV-A6, Table III, Fig. 6).
- Two mode-generation refinements were added: bounded backtracking contact assignment and a temporary per-task robot release when no full-team contact assignment exists. Because both belong to the *shared* mode generator, all policies benefit — at $N=4$ success rises from 70% to **100 %** for PushAround, from 0% to 80% for SL-Push (sim) and from 80% to 90% for DFS-WCCG — as confirmed by instrumentation that counts 6,017 releases and 10,904 backtracking assignments in the baseline logs. PushAround remains the only method that solves all ten trials at every team size, and the most economical (24–30 s planning, 97–110 s execution, 10–11 pushes) versus DFS-WCCG (102–143 s, 172–193 s) and SL-Push (sim) (171–193 s execution,

20–31 pushes). See Sec. III-C2, Sec. IV-A6, Table III, the right half of Fig. 9 (Page 7) of the manuscript, and Fig. 4.

- Sec. III-A2 now derives the criterion from $W/2$ configuration-space inflation with explicit non-convex decomposition, and a new experiment compares WCCG against continuous C-space (CC) and a rasterized reference (FR) on 300 scenes. CC and FR agree in **296/300** cases, whereas WCCG agrees with CC in **300/300** cases and produces **0 FP/FN**. Its mean query time is **1.07** ms, against 1.36 ms for CC and 243.1 ms for FR, i.e. it is **227** $\times$ faster than the raster reference (Table I and Fig. 5).
- Eq. (5) is now presented as the schedule cost guiding the search; the paper states explicitly that PIHS is a feasibility-oriented heuristic search with finite sampling, bounded lookahead, a non-admissible priority, and first-feasible termination, and makes no global optimality or completeness claim. The optimized contact forces are declared feasibility variables (not hardware setpoints), and coupled object–object motion is explained as an outcome of the joint rigid-body rollout in EvalSim (Sec. II-C, II-D, III-C2, III-C3, III-D, III-E1).
- Sec. IV-A1 now states the shared footprints, obstacle configurations, collision/reachability checks, success criterion and simulator, and the shared per-trial budget (**800 s** timeout, **20,000** control steps, **8** parallel workers), with unsuccessful trials assigned the 800 s cap in planning-time statistics. The previous 400 s statement is corrected.

To Reviewer 3

Q: *Overall, this is a good contribution to the field and a well-written paper. The combination of geometric reasoning, recursive pushing, and physics-informed search is interesting, and the real-world validation strengthens the contribution.*

A: We sincerely thank the reviewer for the positive assessment and, in particular, for recognizing the end-to-end integration, recursive pushing, dense-clutter scalability, and hardware validation. The revision addresses the two stated weaknesses and the four follow-up questions without changing the central planning framework. The detailed responses below explain these revisions point by point and provide the exact modified text highlighted in blue.

Q: *The method relies heavily on simulation-based feasibility validation, but sensitivity to uncertainty in mass, friction, force limits, contact dynamics etc. is not systematically evaluated.*

A: We agree that this was the most substantive gap, because physics validation is the distinguishing element of the method. The original hardware experiments demonstrated closed-loop recovery but did not *isolate* sensitivity to individual model parameters. The revision therefore adds a dedicated planner–execution mismatch study with two properties that make the claim falsifiable:

- The planner always uses the nominal model; *only* the execution simulator is perturbed. The experiment therefore measures the consequence of an incorrect planning model rather than the benefit of re-identifying it.

- Obstacle mass, lateral friction, spinning friction and the robot force limit are scaled separately by $\{0.6, 0.8, 1.2, 1.4\}$, i.e. $\pm 20\%$ and $\pm 40\%$, using the same scene seeds as the nominal condition.

Over 17 conditions ($M=30$, $N=2$, 10 seeds each, 170 trials) PushAround succeeds in **166/170 (97.6%)**, and importantly the success rate does *not* degrade with the perturbation magnitude: **78/80 (97.5%)** at $\pm 20\%$ and **78/80 (97.5%)** at $\pm 40\%$, versus 10/10 nominal. The four failures are one mid-execution planning failure (lateral friction $\times 0.6$) and three trials that exceeded the per-trial budget (spin friction $\times 1.2$, force limit $\times 0.6$ and $\times 1.2$). Replanning absorbs the remainder: mean replans rise from 2.3 (nominal) to 3.9 ($\pm 40\%$ average; up to 8.1 under force limit $\times 0.6$), where the weaker pushing force shortens the reachable force range and lengthens execution from 104.7 s to 159.6 s.

Second, to show that this robustness is a property of the method rather than of the scene, we ran SL-Push (sim) and DFS-WCCG under the same protocol (nominal planning, execution-only perturbation, 5 seeds per condition). Under the identical $\pm 40\%$ perturbations, SL-Push (sim) attains **13/40 (32.5%)** — with 0% under lateral friction $\times 1.4$ and force limit $\times 0.6$ — and DFS-WCCG **23/40 (57.5%)**, while requiring $\sim 12\times$ more planning time (mean 229 s versus 18 s). Both baselines are already weaker at nominal physics (3/5 each versus 10/10), which is consistent with the main comparison in Table II.

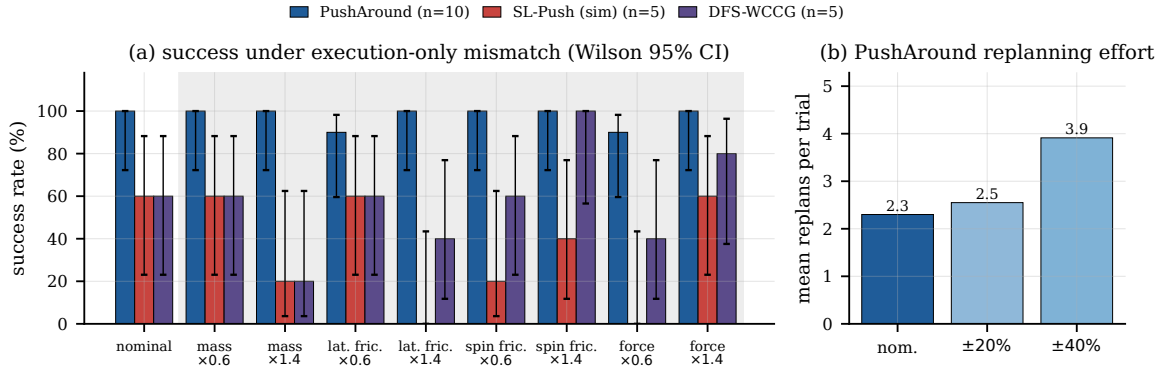


Figure 1: Planner-execution mismatch. (a) Success rate for PushAround ($n=10$ per condition) and for the two baselines ($n=5$) under execution-only perturbations; whiskers are Wilson 95% intervals. (b) Mean number of replans per trial for PushAround, showing that online replanning, rather than a retuned model, absorbs the mismatch.

Changes in the revised manuscript (Page 7, Sec. IV-A6 and Table III; Page 1, Sec. I-B; Page 7, Sec. IV-A6):

- Sec. IV-A6 introduces the mismatch protocol and reports the results; the corresponding rows appear in Table III, and the density/mismatch protocol details are listed in the appendix tables of this letter because the manuscript is page-limited.
- Sec. I-B now states that physics validation is model dependent and that the revision quantifies the sensitivity: **Because physics validation depends on model fidelity, robustness under execution-model mismatch is also evaluated.**
- The claim is stated as: **The planner keeps the nominal model and only the execution simulator is perturbed, one parameter at a time: PushAround succeeds in 156 of 160 perturbed trials (97.5%), the four failures being one low-friction planning failure and three budget timeouts. Under the same $\pm 40\%$ perturbations SL-Push (sim) averages 32.5% and DFS-WCCG 57.5%, at an order of magnitude higher planning time.**

Table 2: Complete planner–execution mismatch grid. PushAround uses 10 seeds per condition; the two baselines 5 seeds, evaluated on the nominal condition and the eight $\pm 40\%$ extremes. PT/ET are means over successful trials, $\bar{R}$ the mean number of replans.

Family	Factor	PushAround ($n=10$)				SL-Push (sim) ($n=5$)			DFS-WCCG ($n=5$)	
		succ.	PT	ET	$\bar{R}$	succ.	PT	ET	succ.	PT
Nominal	1.0	10/10	16.8	104.7	2.3	3/5	33.6	310.9	3/5	156.2
Mass	0.6	10/10	15.9	96.2	2.3	3/5	30.9	300.4	3/5	205.6
Mass	0.8	10/10	15.3	95.0	3.5	—	—	—	—	—
Mass	1.2	10/10	16.3	108.5	2.7	—	—	—	—	—
Mass	1.4	10/10	19.8	124.1	4.4	1/5	40.2	319.3	1/5	136.5
Lateral friction	0.6	9/10	16.5	130.2	3.4	3/5	34.4	315.6	3/5	222.1
Lateral friction	0.8	10/10	14.0	100.0	2.8	—	—	—	—	—
Lateral friction	1.2	10/10	15.8	110.9	2.7	—	—	—	—	—
Lateral friction	1.4	10/10	19.8	116.5	4.0	0/5	—	—	2/5	323.2
Spin friction	0.6	10/10	14.1	91.1	2.7	1/5	28.9	251.4	3/5	179.7
Spin friction	0.8	10/10	15.4	100.9	2.0	—	—	—	—	—
Spin friction	1.2	9/10	14.9	98.9	2.2	—	—	—	—	—
Spin friction	1.4	10/10	16.5	109.2	3.2	2/5	27.3	222.6	5/5	237.0
Force limit	0.6	9/10	29.7	159.6	8.1	0/5	—	—	2/5	360.3
Force limit	0.8	10/10	16.9	121.3	3.0	—	—	—	—	—
Force limit	1.2	9/10	14.7	91.4	1.5	—	—	—	—	—
Force limit	1.4	10/10	15.2	93.3	3.2	3/5	27.6	301.9	4/5	168.1
Aggregate $\pm 20\%$		78/80	15.4	103.4	2.6	—	—	—	—	—
Aggregate $\pm 40\%$		78/80	18.4	115.0	3.9	13/40	31.5	285.2	23/40	229.1
Aggregate all		166/170	16.9	108.9	3.2	16/45	31.8	288.9	26/45	221.0

Q: Although the method is tested with up to 45 movable obstacles, the number of independent trials is limited.

A: We agree, and we have both increased the number of independent runs and changed how the results are reported.

- The density sweep now uses **10 independent seed sequences** per density instead of five (7 densities $\times$ 10 seeds = 70 PushAround trials). For each seed the denser instances are generated incrementally, while different seeds are generated independently, so the success-rate estimates at neighbouring densities are paired but the seeds themselves are not.
- Fig. 10 now shows the success rate and the mean planning time with one-standard-deviation whiskers; Table III lists PT and ET as mean $\pm$ s.d. Failed and pruned baseline trials are assigned the 800 s cap before averaging, and the failure composition (plan failure vs. timeout vs. timeout-pruned) is reported per density in the appendix tables.
- All headline success rates are reported with Wilson 95% intervals in Fig. 6. For PushAround at $M=45$ the interval is $[72.2, 100]\%$ from 10/10; for the baselines at the same density it is $[0, 27.8]\%$ from 0/10.

Changes in the revised manuscript (Page 7, Sec. IV-A6 and Fig. 10 caption; Page 7, Table III):

- Sec. IV-A6 and the caption of Fig. 10 now state the seed count, the nested construction within a seed, and the dispersion: **Obstacle-density scaling in the 8×8 m workspace ($15 \leq M \leq 45$, 10 trials per setting). **Top:** success rate. **Bottom:** mean planning time with failed trials capped at the per-trial 800 s budget; error bars denote one standard deviation.**

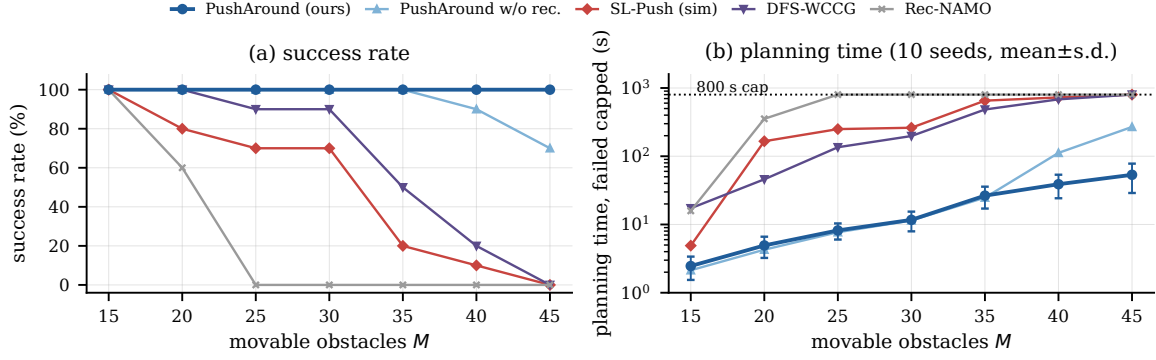


Figure 2: Obstacle-density scaling with 10 independent seeds per density. (a) Success rate; PushAround is the only method that remains at 100% over the whole range. (b) Planning time with failed trials capped at the 800 s budget; whiskers denote one standard deviation over seeds. Rec-NAMO reaches the cap from $M=25$ onward and DFS-WCCG / SL-Push (sim) from $M=45$.

Table 3: Density scaling, success rate (%) and capped mean planning time (s); 7 densities $\times$ 10 seeds per policy. The full table including execution time, push count and failure composition is given in Table 4.

	$M=15$	20	25	30	35	40	45
PushAround (succ.)	100	100	100	100	100	100	100
PushAround (PT)	2.5	4.9	8.2	11.7	26.5	38.9	53.4
PushAround w/o rec. (succ.)	100	100	100	100	100	90	70
PushAround w/o rec. (PT)	2.1	4.3	7.7	11.4	24.7	112.0	268.9
SL-Push (sim) (succ.)	100	80	70	70	20	10	0
SL-Push (sim) (PT)	4.9	165.9	249.9	261.5	652.5	727.0	800.0
DFS-WCCG (succ.)	100	100	90	90	50	20	0
DFS-WCCG (PT)	17.3	45.9	135.5	197.7	483.8	681.7	800.0
Rec-NAMO (succ.)	100	60	0	0	0	0	0
Rec-NAMO (PT)	15.8	353.5	800.0	800.0	800.0	800.0	800.0

- Table III reports PT/ET as $\text{mean} \pm \text{s.d.}$ for the representative densities $M \in \{15, 30, 45\}$.

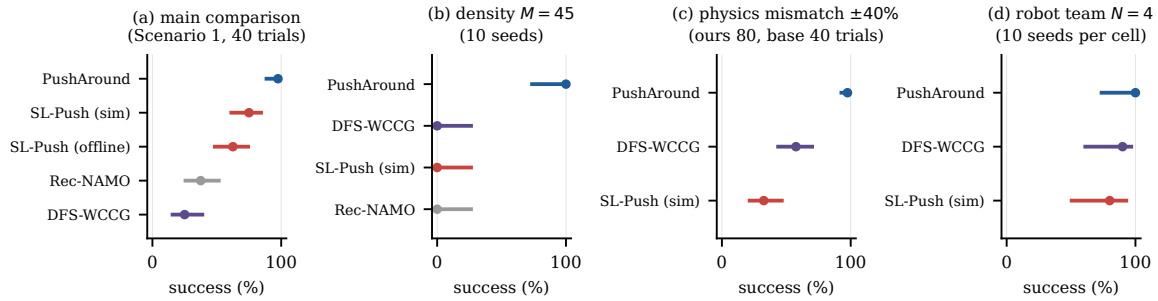


Figure 3: Wilson 95% confidence intervals of the headline success rates. The gaps between PushAround and the baselines are much larger than the interval widths, so the conclusions are stable despite the small per-cell sample sizes (10 seeds in the robot-team study; 10 for PushAround and 5 for the mismatch baseline supplements).

Table 4: Complete obstacle-density grid (10 seeds per cell). Top: success rate (%). Bottom: mean planning time with unsuccessful trials capped at 800 s. PushAround additionally reports execution time and push count.

Success (%)	$M=15$	20	25	30	35	40	45
PushAround	100	100	100	100	100	100	100
PushAround w/o rec.	100	100	100	100	100	90	70
SL-Push (sim)	100	80	70	70	20	10	0
DFS-WCCG	100	100	90	90	50	20	0
Rec-NAMO	100	60	0	0	0	0	0
Planning time (s)	$M=15$	20	25	30	35	40	45
PushAround	2.5 ± 0.9	4.9 ± 1.7	8.2 ± 2.2	11.7 ± 3.7	26.5 ± 9.3	38.9 ± 14.7	53.4 ± 24.5
PushAround w/o rec.	2.1 ± 0.8	4.3 ± 1.3	7.7 ± 2.1	11.4 ± 3.0	24.7 ± 8.2	112.0 ± 242.0	268.9 ± 366.6
SL-Push (sim)	4.9 ± 4.8	165.9 ± 334.2	249.9 ± 379.7	261.5 ± 371.9	652.5 ± 311.5	727.0 ± 230.9	800.0
DFS-WCCG	17.3 ± 9.1	45.9 ± 47.5	135.5 ± 241.1	197.7 ± 248.4	483.8 ± 339.6	681.7 ± 253.5	800.0
Rec-NAMO	15.8 ± 12.4	353.5 ± 385.5	800.0	800.0	800.0	800.0	800.0
PushAround: <i>ET</i> (s)	37.1	56.9	77.3	100.9	165.4	213.2	272.7
PushAround: pushes	3.5	5.8	7.9	10.5	16.6	20.7	26.1

Q: *The experiments primarily use two robots. What happens when the number of robots increases with the complexity of the task? How does the search and the contact/force formulation scale to larger robot teams?*

A: This question turned out to be more productive than expected. It exposed a mode-generation bottleneck, we fixed it, and the re-run — performed on *both* PushAround and the baselines with identical code — shows that the fix lives in the shared mode generator rather than being a PushAround-only behaviour. The corrected comparison is reported below.

(a) Complexity. The implementation never enumerates joint contact tuples. With $|\mathcal{A}|$ reachable contact candidates per robot, exhaustive assignment would require $|\mathcal{A}|^N$ tuples; instead one greedy replacement sweep tests at most $N|\mathcal{A}|$ single-contact substitutions, each solving the force LP whose decision vector has dimension $2N$. Larger teams therefore enlarge the feasible wrench set but also increase contact-assignment and LP cost. Sec. III-E2 states this explicitly.

(b) Bottleneck, fix, and who benefits. The preliminary $N=4$ runs failed in 3 of 10 seeds (2 planning failures, 1 budget timeout). Diagnosis: for some push tasks *no* full-team contact assignment exists — small pushable faces, or the pairwise pusher-clearance constraint rejecting every remaining candidate of one robot — and the legacy greedy replacement was additionally sensitive to robot ordering. We added two mode-generation-local refinements that leave global planning, task sampling and the search tree unchanged: (i) *bounded backtracking* contact assignment and (ii) a *temporary per-task robot release* (up to one robot is excluded for that single push and holds its pose, a penalty is charged during mode ranking so full-team modes are preferred, and the full team is re-available for every later task).

Both refinements sit in the shared mode generator, so the baselines inherit them too — which is why the baseline cells were re-run on the same code. Instrumenting the 40 baseline trial logs (4 cells $\times$ 10 seeds) counts 20,890 failed greedy replacements, 10,904 backtracking assignments (rescuing 739 of them) and 6,017 per-task robot releases. The effect is therefore not PushAround-specific:

After the fix, all three methods solve 80–100% of the team-size trials, so *we no longer claim that the baselines degrade with team size*, and the success-rate differences at $N=3,4$ lie within overlapping Wilson intervals (at $N=4$: PushAround $[0.72, 1.00]$, DFS-WCCG $[0.60, 0.98]$, SL-Push (sim) $[0.49, 0.94]$). What remains separated is (i) *consistency*: PushAround is the only method with 30/30 successes over $N=2,3,4$; and (ii) *efficiency*: 24–30 s of planning and 97–110 s of execution

Table 5: Robot-team scaling at $M=30$ (nominal physics) after the shared mode-generation fix. Every cell uses 10 seeds and the same code; PT/ET are means over successful trials. For reference, the pre-fix history is given in the notes.

Policy	N	Trials	Succ.	PT (s)	ET (s)	pushes	sims
PushAround	2	10	10/10	23.9 ± 10.9	105.0 ± 35.9	11.7	81
	3	10	10/10	30.0 ± 12.7	97.3 ± 42.1	11.1	66
	4	10	10/10	29.2 ± 8.9	110.1 ± 33.1	10.0	59
SL-Push (sim)	3	10	8/10	30.0 ± 46.3	193.3 ± 111.3	31.1	62
SL-Push (sim)	4	10	8/10	23.4 ± 18.2	171.7 ± 61.2	20.5	41
DFS-WCCG	3	10	10/10	142.6 ± 183.1	178.1 ± 72.1	13.9	762
DFS-WCCG	4	10	9/10	101.9 ± 103.9	175.9 ± 113.6	14.3	498

Failure composition: SL-Push (sim) $N=3$ one plan failure and one timeout, $N=4$ two timeouts; DFS-WCCG $N=4$ one timeout; PushAround none.

Pre-fix history (superseded; baselines $n=5$): at $N=3$ PushAround 10/10, SL-Push (sim) 4/5, DFS-WCCG 3/5; at $N=4$ PushAround 7/10, SL-Push (sim) 0/5 (all plan failures), DFS-WCCG 4/5.

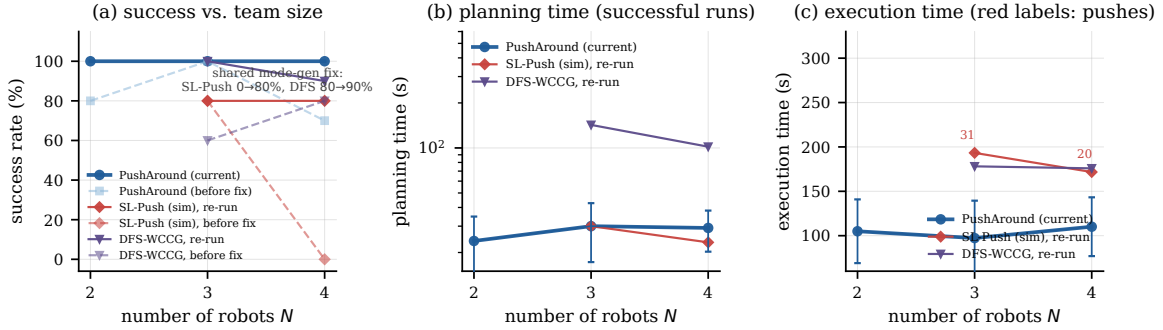


Figure 4: Robot-team scaling after the shared mode-generation fix, with the pre-fix runs shown as dashed history. (a) Success rate: the refinement lifts $N=4$ from 70% to 100% for PushAround, from 0% to 80% for SL-Push (sim) and from 80% to 90% for DFS-WCCG. (b) Planning time and (c) execution time of successful runs; the annotations in (c) are the push counts of SL-Push (sim), which needs twice as many pushes as PushAround.

versus 102–143 s and 172–193 s for DFS-WCCG (whose planning-time standard deviation reaches 183 s at $N=3$), and 10–11 pushes versus 20–31 for SL-Push (sim). The manuscript states this narrower claim rather than a monotone team-size trend.

Changes in the revised manuscript (Page 4, Sec. III-C2; Page 7, Sec. IV-A6 and Table III; Page 5, Sec. III-E2):

- Sec. III-C2 now describes the two refinements: swept over contact slots until J_{MF}^m no longer decreases; a bounded backtracking variant removes the robot-order sensitivity of this replacement. If no full-team assignment exists for a task, e.g., on small faces or under the pairwise pusher-clearance constraint, one robot is released for that single push (charged a penalty when ranking modes, so full-team modes are preferred) and the full team is available again afterwards.
- Sec. IV-A6 and Table III report $N \in \{2, 3, 4\}$; the robot-team group of Table III now lists PushAround and both baseline policies at $N=3$ and $N=4$ (success, PT , ET), so the equal-footing comparison is visible in the table itself rather than only in the text.
- Sec. III-E2 states the $N|\mathcal{A}|$ versus $|\mathcal{A}|^N$ contact-assignment argument.

Table 6: Complete robot-team-scaling grid ($M=30$, nominal physics), all cells on the post-fix code with 10 seeds. The pre-fix rows are kept only as history.

Policy	N	Trials	Succ.	PT (s)	ET (s)	pushes	sims
PushAround (current)	2	10	10/10	23.9 ± 10.9	105.0 ± 35.9	11.7	81
PushAround (current)	3	10	10/10	30.0 ± 12.7	97.3 ± 42.1	11.1	66
PushAround (current)	4	10	10/10	29.2 ± 8.9	110.1 ± 33.1	10.0	59
SL-Push (sim)	3	10	8/10	30.0 ± 46.3	193.3 ± 111.3	31.1	62
SL-Push (sim)	4	10	8/10	23.4 ± 18.2	171.7 ± 61.2	20.5	41
DFS-WCCG	3	10	10/10	142.6 ± 183.1	178.1 ± 72.1	13.9	762
DFS-WCCG	4	10	9/10	101.9 ± 103.9	175.9 ± 113.6	14.3	498
<i>Pre-fix history (superseded; baselines $n=5$)</i>							
PushAround	2	10	8/10	19.5	125.6	—	—
PushAround	3	10	10/10	21.8	100.6	—	—
PushAround	4	10	7/10	23.3	121.3	—	—
SL-Push (sim)	3	5	4/5	61.7	196.7	—	—
SL-Push (sim)	4	5	0/5	—	—	—	—
DFS-WCCG	3	5	3/5	164.6	199.8	—	—
DFS-WCCG	4	5	4/5	131.0	190.9	—	—

Q: What guarantees, if any, does the hybrid search provide regarding solution quality?

A: Thank you for insisting on this. No global optimality or completeness guarantee is claimed, and the revision now says so explicitly in three places rather than leaving it implicit.

PIHS samples a finite set of pushing directions and contact modes, uses bounded gap lookahead, ranks nodes with a heuristic best-first priority whose remaining-cost term is *not* required to be admissible, and terminates at the first physics-validated W -clear solution. The returned schedule is therefore a feasible schedule, not a globally minimal one. Consistently, Eq. (5) is presented as the clearance cost that ranks candidate schedules; it is also stated that the objective couples obstacle selection, mode generation, and switching into a hybrid combinatorial–continuous problem that the implemented search does not solve to global optimality.

Changes in the revised manuscript (Page 2, Sec. II-D; Page 4, Sec. III-D; Page 5, Sec. III-D, node selection; Page 5, Sec. III-D, termination):

- Equation (5) ranks feasible schedules; PIHS uses finite sampling and does not guarantee global optimality of the continuous hybrid objective.
- The search operates on a finite sampled candidate set and therefore does not guarantee global optimality or completeness of the underlying continuous hybrid planning problem.
- The remaining-cost term is a heuristic and is not required to be admissible. The score prioritizes nodes by realized cost and promising future actions, but it does not provide an A* optimality guarantee.
- The returned schedule is described as a feasible solution satisfying the terminal clearance condition and validated transitions, without implying global minimality of $\mathcal{J}_{\text{clr}}(\pi)$.

Q: Is the time needed for the final vehicle to actually traverse the cleared corridor included in this objective, or is the objective only concerned with clearing the obstacles? Similarly, does J only capture the robots’ effort in moving the obstacles?

A: The reviewer’s reading is exactly the intended one, and the manuscript now states both points explicitly for the same reason.

- $T \triangleq \sum_{k=1}^K \Delta t_k$ is the duration of the robotic clearance operation *only*. The traversal of the external vehicle is not included in T ; it enters the formulation solely as the terminal W -clearance condition.
- $J(\cdot)$ is the simulation-estimated manipulation effort of the robots when moving obstacles. It does not model vehicle traversal cost. In addition, the vehicle does not move concurrently with the clearing operation: it traverses the corridor only after the robots have finished clearing, which is why traversal time plays no role in the objective and appears only in the terminal W -clearance condition.

Changes in the revised manuscript: Sec. II-D now reads *where $T \triangleq \sum_{k=1}^K \Delta t_k$ denotes the robotic clearance duration excluding subsequent vehicle traversal, and $J(\cdot)$ is the simulation-estimated manipulation effort. Vehicle traversal is considered only through the terminal W -clearance condition.*

Q: *The paper demonstrates online replanning when the real execution differs from the simulation. How can simple pushing robots evaluate this discrepancy?*

A: We agree that the previous wording (“the realized motion deviates from simulation”) was too vague and could be read as requiring online parameter identification or force sensing. That is not the implementation, and the revision replaces the phrase with an explicit state-based monitor.

In the hardware experiments, motion capture provides robot and object poses. A push is declared to deviate — and replanning is triggered from the observed state — when the measured object pose departs from the successor state predicted by EvalSim by more than **0.3 m in position** or $\pi/4$ **in orientation**, or when the transition/contact checks fail (blocked pre-contact transition, contact loss or stall, unsatisfied geometric task predicate, invalidated WCCG connectivity for the remaining schedule). The pose threshold is deliberately loose: a slightly larger or slightly off-axis push is accepted rather than aborted, because such deviations are routinely recoverable by the subsequent tasks. No onboard force sensing or mass/friction estimation is required.

Changes in the revised manuscript: Sec. IV-B1 now states *Motion capture provides the current robot and object poses, which are compared with the successor state predicted by EvalSim: a push is deemed to deviate and triggers replanning only when the measured object position error exceeds 0.3 m or its orientation error exceeds $\pi/4$. The threshold is deliberately loose, tolerating a slight over-push or off-axis push.* Sec. III-E1 additionally lists the non-pose triggers.

To Reviewer 4

Q: Overall, I think the approach is technically reasonable and the experiments support the proposed method. My comments are mainly about clarification rather than major technical concerns, and I would support acceptance after these revisions.

A: We thank the reviewer for the supportive assessment and for pinpointing four places where the presentation was under-specified. Each is addressed below with the corresponding revised text.

Q: The WCCG connectivity condition is important since it is used throughout the search and also determines termination. The intuition is reasonably clear, but the paper would benefit from a little more justification here, or perhaps a comparison with a standard configuration-space connectivity check. This seems especially relevant when the obstacles are non-convex.

A: We agree, and we provide both a derivation and the suggested empirical check.

(a) Why the same-face test is the right query. For a vehicle whose footprint is enclosed by a disk of radius $W/2$, inflating every obstacle boundary by $W/2$ turns any pair of boundary components whose closest distance is below W into *overlapping* barriers in the configuration space of the vehicle centre. A WCCG bridge–bridge edge records exactly such a non-traversable bottleneck, and centroid–bridge edges attach the bottleneck to the adjacent free regions. Consequently, “start and goal lie in the same planar face” is equivalent to connectivity in the complement of the inflated geometry, i.e. to the W -clearance condition. Non-convex obstacles require care, which is why the construction first performs a clearance-relevant convex decomposition and then inserts bridges between the resulting components/segments.

(b) Empirical validation on 300 scenes. Because degenerate geometry and decomposition choices still matter in practice, we also compare WCCG against two independent references on 300 random scenes stratified into convex, mixed, and non-convex obstacles: a continuous configuration-space (CC) reference and a rasterized reference (FR) at 0.02 m, verified at 0.01 m. CC and FR disagree in **4/300** cases, all of which are raster-unstable near narrow or tangential contacts. On all 300 scenes, WCCG agrees with the continuous reference and produces **zero false positives and zero false negatives** — a false positive would be the dangerous case, as it could terminate PIHS prematurely. WCCG additionally answers a query in **1.07 ms** on average, comparable to CC (1.36 ms) and **227×** faster than FR (243.1 ms), while supplying the gap graph that PIHS consumes.

Table 7: WCCG connectivity validation on 300 random scenes (10 movable obstacles, 8×8 m, clearance $W=1$ m). FP/FN are WCCG errors with respect to the continuous reference. Times are means over 100 scenes per category.

Category	CC=FR	WCCG vs. CC			Time (ms)			FR/WCCG
		agree	agree	FP FN	CC	FR	WCCG	
Convex	100/100	100/100	0	0	1.23 ± 0.13	282.9 ± 52.3	0.86 ± 0.15	330
Mixed	98/100	100/100	0	0	1.33 ± 0.13	242.4 ± 55.7	1.05 ± 0.17	231
Non-convex	98/100	100/100	0	0	1.51 ± 0.18	204.1 ± 51.4	1.30 ± 0.19	156
Total	296/300	300/300	0	0	1.36 ± 0.19	243.1 ± 62.0	1.07 ± 0.25	227

Changes in the revised manuscript (Page 3, Sec. III-A2; Page 6, Sec. IV-A3 and Table I; Page 3, Sec. III-A):

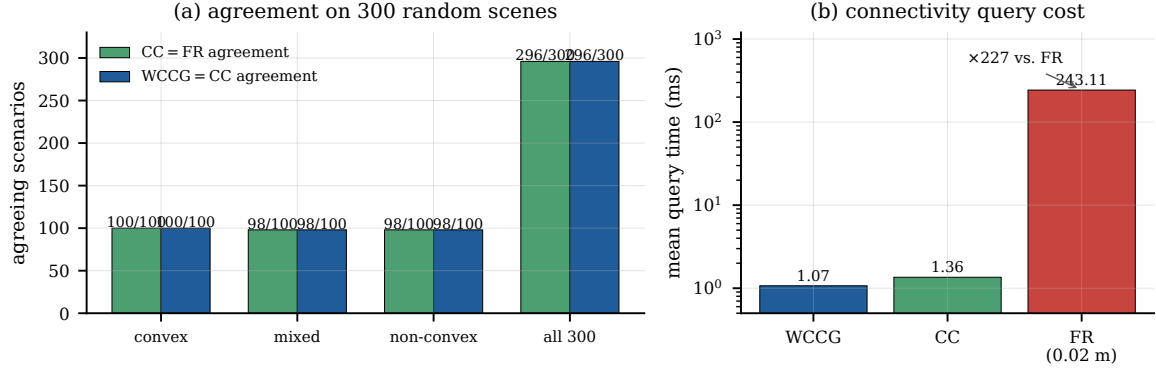


Figure 5: WCCG validation. (a) Agreement per category: CC and FR differ on 4 of 300 scenes (all raster-unstable), whereas WCCG matches the continuous reference on all 300 with zero false positives/negatives. (b) Mean connectivity-query cost on a logarithmic scale.

- The criterion follows the standard configuration-space inflation interpretation: expanding obstacles by a disk of radius $W/2$ turns gaps narrower than W into overlapping barriers. Non-convex obstacles are handled through clearance-relevant decomposition, after which WCCG records the separating bottlenecks. The consistency between WCCG connectivity and continuous and rasterized configuration-space references is further evaluated in Sec. IV-A3.
- Sec. IV-A3 and Table I report the validation above; the runtime columns are now backed by the released timing logs rather than by an earlier statement, and the table footnote states the measurement scope — CC, FR and WCCG are timed in one harness over the same 300 scenes, with the WCCG query served by its compiled backend.
- The absolute phrase “eliminates resolution-induced errors” was replaced by an accurate statement: WCCG avoids introducing a workspace-grid resolution into the repeated queries, while still depending on a correct decomposition and bridge construction.

Q: Eq. (5) is written as an optimization problem, while the implementation uses sampled pushing directions, limited lookahead, and a heuristic best-first search that terminates once a feasible solution is found. The authors should clarify whether any optimality or completeness claim is intended. If not, it may be better to describe Eq. (5) as the objective used to guide the search.

A: We agree completely with this reading and have adopted the reviewer’s suggested framing. Eq. (5) is now introduced as the clearance cost $\mathcal{J}_{\text{clr}}(\pi)$ that ranks candidate schedules. The text distinguishes three levels explicitly: (i) the underlying continuous hybrid problem, (ii) the finite candidate space actually sampled by PIHS, and (iii) the implemented algorithm — a feasibility-oriented heuristic best-first search that stops at the first validated W -clear solution. No optimality or completeness claim is made at any level. This is now stated in Sec. II-D (Page 3), in the node-selection discussion (Page 5, where the remaining-cost term is explicitly not required to be admissible) and in the termination discussion (Page 5).

Q: It is not completely clear how the optimized contact forces are used on the hardware. The pushing mode includes both contact points and forces, but the execution section mainly describes

moving the robots to the contact configuration and then following the desired pushing direction. It would be useful to briefly explain the low-level controller and whether these forces are actually commanded or are mainly used to check feasibility. Related to this, I am also not sure whether coupled motion between multiple objects, such as in the second hardware experiment, is explicitly considered during planning or is mostly handled by the physics simulation.

A: Thank you for raising both points; we agree that the interface between the optimized forces and the actual hardware commands was not made explicit, and we have rewritten the relevant paragraph accordingly.

For a candidate contact tuple, the LP decides whether the robots can generate a wrench compatible with the desired object motion within the modelled force limits, friction cone, and unilateral contact constraints. The optimized vector is therefore used to *certify and rank* wrench feasibility; it is not sent to the hardware. The Mecanum UGVs receive planar velocity commands (`geometry_msgs/Twist`) and track the mode-defined contact poses. The controller also follows a *short forward object-motion reference*: the object displacement and rotation predicted by EvalSim over the next control interval (a few centimetres and a few degrees), which the velocity commands are commanded to realize. The optimized force therefore never reaches the actuators — it only determines whether such a reference lies inside the feasible wrench set, and the reference itself is what the feedback controller tracks, with motion-capture feedback and replanning absorbing the difference.

Coupled motion is instead an outcome of the rollout. A high-level strategy names one directly targeted obstacle, but EvalSim advances all robots and all movable bodies jointly; object–object contacts are resolved by the physics engine and the complete successor scene is returned to the search. The adjacent T-shaped block moved in hardware Scenario 2 is exactly this contact-induced motion, not a separately scripted task — which is also why the revision keeps the coupled rollout as the source of physical validation.

Changes in the revised manuscript (Page 2, Sec. II-C; Page 4, Sec. III-C3; Pages 5 and 7, Secs. III-E1 and IV-B1; Page 7, Sec. IV-B2; Pages 4 and 6, Secs. III-C2 and IV-A1; Page 6, Sec. IV-A1; Pages 6–7, Secs. IV-A1 and IV-A6):

- The force vector $\mathbf{u}_m$ serves as a feasibility variable to certify wrench generation under the modeled contact and actuation constraints rather than a directly commanded control input.
- The rollout jointly simulates all robots and movable bodies, thereby capturing contact-induced object interactions. and $\mathbf{u}_m^k$ certifies wrench feasibility while hardware tracks the contact geometry from pose/velocity commands.
- Sec. III-E1 and Sec. IV-B1 describe the velocity-level controller and the motion-capture feedback loop.
- This neighboring-object motion is not a separately encoded action but an emergent outcome of the coupled rigid-body rollout in EvalSim, whose snapshot is used to update the WCCG and continue replanning.

Q: A few more implementation details, such as robot reachability checks, timeout limits, and computation budget, would make the comparison easier to interpret. I would also suggest adding

error bars or standard deviations, particularly for the scalability experiment where only five seeds are used for each obstacle density.

A: We have made the protocol explicit and now report dispersion throughout. The full protocol is summarized in Table 8.

- A contact candidate is retained only if the associated pre-contact pose is collision-free and the transition planner can reach it from the current robot configuration; a multi-robot tuple is rejected if any assigned robot fails this check. All compared methods use this same check, together with identical footprints, obstacle realizations, success criterion, and execution simulator.
- All suites use a per-trial timeout of **800 s**, a cap of **20,000** control steps, and **8** parallel workers; plan failures, timeouts and timeout-pruned trials are assigned the 800 s value before averaging planning times, and the failure composition is reported per condition.
- The density sweep, the robot-team study (all policies) and the mismatch study for PushAround use 10 seeds per cell; only the *mismatch* baseline supplements use 5 seeds and are labelled as such. Fig. 6 reports Wilson 95% intervals for every headline number, and Table III in the manuscript reports $\text{mean} \pm \text{s.d.}$
- The original manuscript stated a 400 s cap; the stored logs use 800 s, and the revision corrects this.

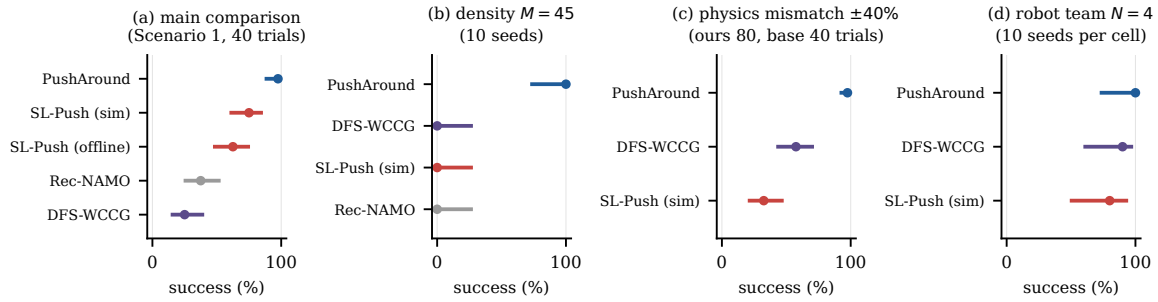


Figure 6: Wilson 95% confidence intervals of the headline success rates. The gaps between PushAround and the baselines are much larger than the interval widths, so the conclusions are stable despite the small per-cell sample sizes (10 seeds in the robot-team study; 10 for PushAround and 5 for the mismatch baseline supplements).

Changes in the revised manuscript: Sec. IV-A1 now states [All methods use the same robot footprints, obstacle configurations, collision/reachability checks, and execution simulator, and are given the same per-trial timeout of 800 s and planning budget of 20,000 control steps; unsuccessful trials are assigned the 800 s cap in the planning-time statistics.](#)

Table 8: Protocol and budget summary for all reported studies. All suites share the simulator, the success criterion, and the collision/reachability checks.

Study	Workspace	M	N	Seeds	Timeout / cap	Workers
Main comparison (Table II)	8×5 m	10	2	40 trials	800 s / 20k	8
Density (Table III, Fig. 10)	8×8 m	15–45	2	10 per density	800 s / 20k	8
Mismatch (Table III)	8×8 m	30	2	10 (ours) / 5 (base)	800 s / 20k	8
Robot scaling (Table III)	8×8 m	30	2–4	10 (ours) / 5 (base)	800 s / 20k	8
WCCG validation (Table I)	8×8 m	10	—	300 scenes	n/a	4 (C++)

Scene generation in the generalization studies uses movable-object masses sampled from $[20, 30]$ kg; the Scenario-1 comparison uses $[5, 15]$ kg. PushAround’s density results are produced with a precomputed mode table (`use_mode_table`), whereas the mismatch and robot-scaling studies use greedy mode optimization; this explains the small planning-time differences between the corresponding nominal rows (e.g. 11.7 s versus 16.8 s at $M=30$, $N=2$) and does not affect success rates.

Updated, added and removed figures

Three figures of the manuscript changed in this revision. In each pair the left image is the original version (first submission) and the right image is the revised version; the page refers to the location in the revised manuscript.

F1. Obstacle-density scaling: revised Fig. 11 (Page 7)

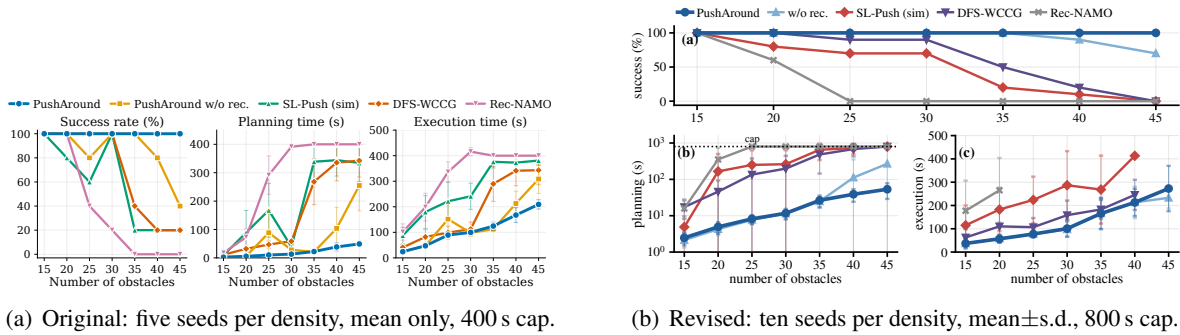


Figure 7: Obstacle-density scaling, original (left) versus revised (right). The panel structure is unchanged, so the two images are directly comparable: the revision doubles the number of independent seeds per density, adds one-standard-deviation error bars to both timing panels, and caps unsuccessful trials at the 800 s budget instead of 400 s, so failed runs are no longer displayed as favourable planning times.

F2. Simulation snapshots: revised Fig. 8 (Page 6)

F3. Scalability scenes: revised Fig. 10 (Page 7)

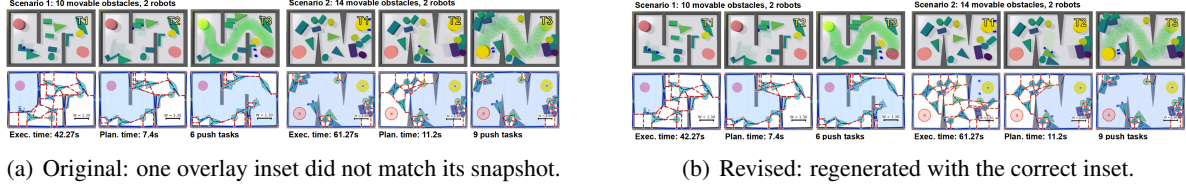


Figure 8: Execution and planning alignment in the two simulation scenarios, original (left) versus revised (right). One inset of the original composite was incorrect, because the fourth sub-image of its bottom row duplicated the sixth sub-image of the same row, so that WCCG overlay did not correspond to the snapshot above it. The figure has been regenerated with the correct panel; the caption, the scenario and the message are unchanged.

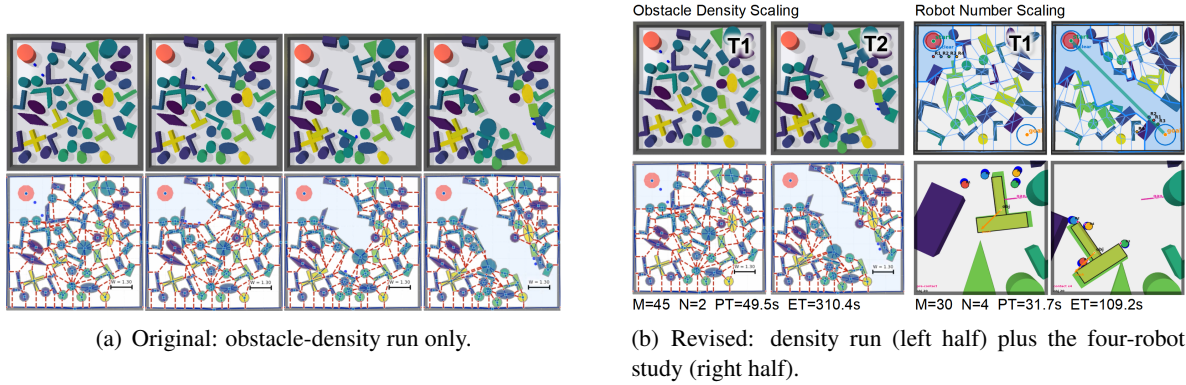


Figure 9: Simulation scenes of the scalability evaluation, original (left) versus revised (right). The revised figure covers both scalability axes instead of the obstacle-density axis alone: its left half keeps the representative $M=45$, two-robot run (snapshots above, WCCG overlays with the start face in blue below), while its right half adds the four-robot study ($M=30$, $N=4$) with the global initial and final states above and two magnified views of a single pushing task below. This supplies visual evidence for the team-size results of Table III (Page 7), which previously had no accompanying figure.

PushAround: Collaborative Path Clearing via Physics-Informed Hybrid Search

Abstract—The passage of large vehicles in cluttered environments is often blocked by movable obstacles, motivating the use of mobile robot teams to proactively clear traversable corridors. Existing approaches to Navigation Among Movable Obstacles (NAMO) mainly plan abstract sequences of obstacle displacements but often neglect physical feasibility, overlooking robot dimensions, obstacle mass, contact geometry, and required forces, which can lead to strategies that fail in execution. This work introduces *PushAround*, a physics-informed hybrid search framework that incrementally constructs executable clearing plans by coupling geometric connectivity reasoning with short-horizon simulation. Repeated W-clearance connectivity graph (WCCG) queries expose frontier gaps where the vehicle corridor must be widened and rank them by estimated downstream costs, while the search generates direct or recursive pushing candidates – choosing which obstacles to displace, in which directions, and with which contact points and forces – admitting only physics-validated candidates. **Additional validation characterizes connectivity-query consistency and sensitivity to physics-model mismatch.** The framework is validated by extensive simulations and hardware experiments.

I. INTRODUCTION

In many real-world settings, the passage of large vehicles or transport units is obstructed by movable obstacles such as pallets, boxes, or equipment, motivating the deployment of mobile robot teams to actively clear traversable corridors and enable safe passage. This capability is critical in cluttered and unstructured environments such as warehouses, disaster sites, and dense urban spaces. Conventional path planning methods assume static obstacles and thus fail once direct routes are blocked [1]. Navigation Among Movable Obstacles (NAMO) has been studied as a means of creating new routes by pushing, pulling, or rotating obstacles [2], yet most methods abstract away physical feasibility. Factors such as robot dimensions, obstacle size and mass, contact geometry, applied forces, and the coupled dynamics of pushing are typically ignored, producing plans that are *geometrically valid but physically unrealizable*. As shown in Fig. 1, this difficulty is amplified when a large robot cannot reach obstacle boundaries, while a small robot lacks sufficient pushing force. Cooperative pushing by several small robots is a viable alternative, but introduces strong physical coupling since simultaneous pushes can trigger chained motions of multiple obstacles, making prediction and planning substantially more complex.

A. Related Work

NAMO extends motion planning to cluttered environments by planning obstacle displacements through pushing, pulling, or rotating [2], [3]. Later work considers heuristic search, learning-based local planning, physics-based rollouts, and collaborative pushing [3]–[7]. However, many NAMO formulations simplify robot geometry, actuation limits, object mass, and contact dynamics, which can yield unrealizable plans.



Fig. 1. Collaborative path clearing in both simulation (**Top**) and hardware (**Bottom**). Two robots actively push movable obstacles to create a W-clear path for a large vehicle to traverse from start to goal, via a hybrid search over the sequence of obstacles to push and the associated pushing modes.

Collaborative pushing has been widely studied in multi-robot manipulation: foundational work on the mechanics of pushing and limit surfaces [8], [9], cooperative transport with robot swarms [10], multi-robot box pushing with reinforcement learning [11], collaborative pushing of polytopic objects in cluttered scenes [7], and learning-based locomanipulation [12]. These studies focus on a prescribed object rather than multi-object corridor clearing.

Physics-informed planning incorporates dynamics models, contact reasoning, or simulation feedback into motion generation: learned dynamics for humanoid contact planning [13], multi-contact force control [14], demonstration-guided optimal control over contact modes [15], physics simulation interleaved with multi-agent pathfinding [16], [17], and learned physical priors [18], [19]. **These methods nonetheless assume a prescribed manipulation target or few objects, and physical realizability is usually argued rather than validated before execution. Corridor clearing differs on both counts: many movable obstacles are rearranged jointly, and the cleared configuration must remain traversable for a larger vehicle.**

B. Our Method

As shown in Fig. 2, this work introduces **PushAround**, a physics-informed framework for collaborative multi-robot pushing that actively constructs traversable corridors for large vehicles in cluttered environments. Its central novelty is a hybrid search that jointly determines which obstacles to displace and how to push them, coupling the high-level sequence of obstacle clearing with the low-level pushing modes that specify contact points, directions, and forces. The framework integrates three components: a W-clearance connectivity graph (WCCG) that certifies vehicle passage under width constraints, a gap-ranking strategy that prioritizes frontier gaps by estimated effort, and pushing-mode generation under geometric constraints. These are encapsulated in a search that incrementally expands candidate clearing plans while validating

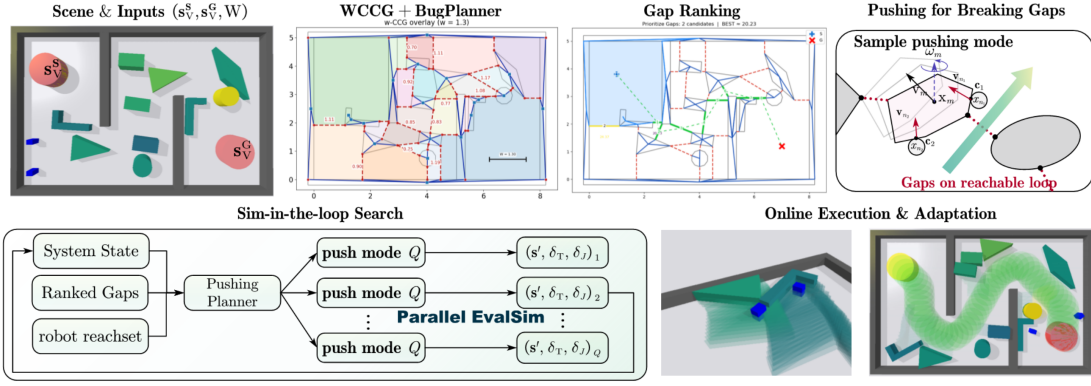


Fig. 2. Overview of the proposed framework. **Top:** Scene inputs, WCCG construction, gap ranking, and sampling of pushing modes for breaking frontier gaps; **Bottom-left:** Sim-in-the-loop hybrid search with parallel validation in simulation; **Bottom-right:** Online execution and adaptation during path clearing.

their physical realizability, so the resulting plans are not only geometrically valid but also feasible under robot dimensions, object mass, contact geometry, and coupled pushing dynamics. **Since physics validation depends on model fidelity, robustness under execution-model mismatch is also evaluated.**

The contributions are twofold: (I) it introduces a unified framework that couples collaborative pushing with physics-informed verification, producing executable plans for path clearing in cluttered environments; and (II) it shows significant improvements in feasibility, efficiency and scalability over existing NAMO and collaborative clearing methods.

II. PROBLEM DESCRIPTION

A. Workspace and Robots

The workspace is a bounded planar region $\mathcal{W} \subset \mathbb{R}^2$ containing immovable structures $\mathcal{O} \subset \mathcal{W}$ and $M \geq 0$ movable rigid polygons $\Omega \triangleq \{\Omega_1, \dots, \Omega_M\} \subset \mathcal{W}$. Each movable obstacle Ω_m has mass M_m , inertia I_m , frictional parameters and state $\mathbf{s}_m(t) \triangleq (\mathbf{x}_m(t), \psi_m(t))$, with $\mathbf{x}_m(t) \in \mathbb{R}^2$ the centroid position and $\psi_m(t) \in \mathbb{R}$ the orientation; $\Omega_m(t)$ denotes the region it occupies at time t . A team of N robots, $\mathcal{R} \triangleq \{R_1, \dots, R_N\}$, operates as a cooperative unit. Each robot R_i is a rigid body with state $\mathbf{s}_{R_i}(t) \triangleq (\mathbf{x}_{R_i}(t), \psi_{R_i}(t))$ and footprint $R_i(t)$. The instantaneous free space is given by:

$$\widehat{\mathcal{W}}(t) \triangleq \mathcal{W} \setminus \left(\mathcal{O} \cup \{\Omega_m(t)\}_{m=1}^M \cup \{R_i(t)\}_{i=1}^N \right), \quad (1)$$

where $\widehat{\mathcal{W}}(t)$ excludes regions occupied by immovable obstacles, movable obstacles, and robots.

B. External Vehicle and Clearance Goal

An external vehicle $\mathcal{V}$ whose footprint is enclosed by a disk of radius $W/2 > 0$ must navigate within the workspace from a start position $\mathbf{x}_V^S \in \mathcal{W}$ to a goal position $\mathbf{x}_V^G \in \mathcal{W}$. Since movable obstacles may obstruct the way, the clearance goal is to rearrange them so that the workspace contains a passage from $\mathbf{x}_V^S$ to $\mathbf{x}_V^G$ that can accommodate this enclosing disk, i.e., a passage of width at least W . At time t , this is expressed by the admissible reference-center space $\mathcal{F}_W(t) \triangleq \{p \in \widehat{\mathcal{W}}(t) \mid \mathbb{B}_{W/2}(p) \subset \widehat{\mathcal{W}}(t)\}$, where $\mathbb{B}_{W/2}(p)$ is the disk of radius $W/2$ centered at p . The W -clearance condition is then given by:

$$\exists \mathcal{P}_V^W \subset \mathcal{F}_W(t) : \mathbf{x}_V^S \rightsquigarrow \mathbf{x}_V^G, \quad (2)$$

which certifies a collision-free centerline path for any vehicle footprint contained in the enclosing disk.

C. Collaborative Pushing Modes

To enlarge blocked gaps, the robots may actively reconfigure Ω by pushing movable obstacles. For obstacle $\Omega_m \in \Omega$, a pushing mode is denoted by $\xi_m \triangleq (\mathbf{c}_m, \mathbf{u}_m)$, where $\mathbf{c}_m = (c_1, \dots, c_N)$ is an ordered tuple of contact points on $\partial\Omega_m$, and the ordering specifies the robot-to-contact assignment. The vector $\mathbf{u}_m \in \mathbb{R}^{2N}$ stacks the corresponding planar contact forces, expressed in the body frame of Ω_m . For a contact tuple $\mathbf{c}_m$, the set of admissible forces $U_m(\mathbf{c}_m) \subset \mathbb{R}^{2N}$ collects contact forces satisfying the robot pushing limits, robot-object friction, and unilateral contact constraints. The body wrench induced by a mode is defined as:

$$\mathbf{q}_m(\xi_m) \triangleq G_m(\mathbf{c}_m)\mathbf{u}_m, \quad (3)$$

where $G_m(\mathbf{c}_m) \in \mathbb{R}^{3 \times 2N}$ is the contact-to-wrench mapping; and $\mathbf{q}_m(\xi_m) \in \mathbb{R}^3$. The admissible mode set is state dependent as denoted by $\Xi_m(\mathbf{s})$. It contains modes whose contacts are reachable from the current robot configurations, pre-contact configurations are feasible in the current clutter, and forces satisfy $\mathbf{u}_m \in U_m(\mathbf{c}_m)$. Hence, applying a feasible mode $\xi_m \in \Xi_m(\mathbf{s}(t))$ yields the physical transition:

$$\mathbf{s}(t^+) \triangleq \Phi(\mathbf{s}(t), m, \xi_m), \quad (4)$$

where $\mathbf{s}(t)$ stacks all robot and obstacle states. **The force vector $\mathbf{u}_m$ serves as a feasibility variable to certify wrench generation under the modeled contact and actuation constraints rather than a directly commanded control input.**

D. Problem Statement

The goal is to compute a *hybrid schedule* that reconfigures movable obstacles to admit a W -feasible path for the external vehicle. The schedule is defined as:

$$\pi \triangleq \{(\Omega_{m_k}, \xi_k, \Delta t_k)\}_{k=1}^K, \quad (5)$$

where $\Omega_{m_k} \in \Omega$ is the manipulated obstacle, $\xi_k \in \Xi_{m_k}$ is its pushing mode, and $\Delta t_k > 0$ is the execution duration. The search is guided by

$$\mathcal{J}_{\text{clr}}(\pi) \triangleq T + \alpha \sum_{k=1}^K J(m_k, \xi_k, \mathbf{s}(t_k)), \quad (6)$$

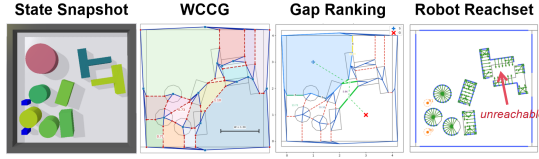


Fig. 3. Core components of the physics-informed hybrid search: WCCG construction, gap ranking, mode generation, and physics validation.

where $T \triangleq \sum_{k=1}^K \Delta t_k$ denotes the robotic clearance duration excluding subsequent vehicle traversal, and $J(\cdot)$ is the simulation-estimated manipulation effort. Vehicle traversal is considered only through the terminal W -clearance condition (2). The constraints (1)–(4) enforce valid pushing transitions. Equation (6) ranks feasible schedules; PIHS uses finite sampling and does not guarantee global optimality of the continuous hybrid objective.

Remark 1. The objective and constraints above couple obstacle selection, pushing-mode generation, and switching in one hybrid planning problem, yielding a combinatorial-continuous search space far richer than classical NAMO formulations with simple objects or hand-crafted contact modes [8], [10], [11]. ■

III. PROPOSED SOLUTION

As shown in Fig. 2, PushAround first uses the W -Clearance Connectivity Graph in Sec. III-A and gap ranking in Sec. III-B to identify and prioritize blocking gaps. It then generates and evaluates physically feasible collaborative pushing modes in Sec. III-C. These components are integrated into the simulation-in-the-loop hybrid search in Sec. III-D, followed by the online execution procedure summarized in Sec. III-E.

A. W -Clearance Connectivity Graph

The feasibility of routing the vehicle reduces to whether a W -width corridor connects $\mathbf{x}_v^S$ and $\mathbf{x}_v^G$. A W -clearance connectivity graph (WCCG) is introduced here to encode obstacle adjacencies under clearance W and enable fast connectivity queries. Conventional clearance checks either rasterize the workspace into an inflated occupancy grid or query sampling-based roadmaps, and must therefore fix a resolution and re-query the map whenever an obstacle moves [20]. Built directly from obstacle geometry, the WCCG introduces no grid-resolution parameter in the repeated connectivity queries used during planning.

1) *Graph Construction:* The WCCG is built by decomposing all clearance-relevant obstacle and workspace boundaries into convex components or segments. From each component C , a centroid node v_c is created. When two components C_u and C_v have closest points $p_u \in C_u$ and $p_v \in C_v$ with distance less than W , bridge nodes are added at p_u and p_v . These bridge nodes are connected by a bridge-bridge edge, annotated with the gap width $w_{uv} \triangleq \|p_u - p_v\|$, and further linked back to their centroids with centroid-bridge edges. Narrow passages with $w_{uv} < W$ are marked as potential bottlenecks. The resulting graph is given by:

$$\mathcal{G}_W \triangleq (\mathcal{V}, \mathcal{E}_c \cup \mathcal{E}_b), \quad (7)$$

where $\mathcal{V}$ are centroid and bridge nodes; $\mathcal{E}_c$ contains centroid-bridge edges; and $\mathcal{E}_b$ for bridge-bridge edges with widths w_{uv} .

2) *Connectivity Criterion:* Once $\mathcal{G}_W$ has been constructed, connectivity queries can be performed without explicitly computing a geometric path. Let $\text{Face}(\mathcal{G}_W)$ denote the set of connected planar faces induced by the embedded WCCG, i.e., the maximal regions separated by obstacle components and WCCG bottleneck edges. The criterion for (2) is given by:

$$(\mathbf{x}_v^S, \mathbf{x}_v^G \in \mathcal{F}_W(t)) \wedge (\exists F \in \text{Face}(\mathcal{G}_W) : \mathbf{x}_v^S, \mathbf{x}_v^G \in F) \quad (8)$$

The criterion follows the standard configuration-space inflation interpretation: expanding obstacles by a disk of radius $W/2$ turns gaps narrower than W into overlapping barriers. Non-convex obstacles are handled through clearance-relevant decomposition, after which WCCG records the separating bottlenecks. The consistency between WCCG connectivity and continuous and rasterized configuration-space references is further evaluated numerically in Sec. IV-A3. A frontier-tracing procedure, similar to the BugPlanner [21], is then used to extract a skeleton Σ as an ordered sequence of centroid and bridge nodes. Starting from $\mathbf{x}_v^S$, it casts a ray toward $\mathbf{x}_v^G$ and follows the encountered frontier loop until either a valid exit is found or a blocking cycle is returned.

B. Ranking of Frontier Gaps

When the condition in (8) fails, the vehicle cannot reach $\mathbf{x}_v^G$ from $\mathbf{x}_v^S$ through a W -clear path $\mathcal{P}_W^G$. In this case, the planner must prioritize *blocking gaps* on the reachable frontier of $\mathcal{G}_W$ according to their predicted downstream cost, and feed the ranked list to the hybrid search as local clearance targets.

1) *Frontier Extraction and Candidate Gaps:* A BugPlanner query on $\mathcal{G}_W$ above either confirms connectivity or returns a counter-clockwise frontier loop $\mathcal{L}$ that separates the two endpoints. The bridge-bridge edges on $\mathcal{L}$ define the first-hop candidate set $\Gamma_{\mathcal{L}} \triangleq \{g_1, \dots, g_K\}$, where each gap g_k is represented by its two closest boundary points, its width $w_{g_k} < W$, and the adjacent obstacles that can potentially be moved. These candidates are then ranked by predicted downstream cost and passed to the hybrid search as local clearance targets.

2) *Evaluation for Immediate Cost:* Each candidate $g \in \Gamma_{\mathcal{L}}$ is evaluated by combining the cost for the robots to reach the gap and the effort required to widen it. Let $\mathbf{s}_{\mathcal{R}}$ be the current robot positions, and $\mathbf{o}_g$ as the outside insertion point of gap g . The resulting one-hop cost is given by:

$$\mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) = \lambda_t \mathcal{C}_t(\mathbf{s}_{\mathcal{R}}, \mathbf{o}_g) + \lambda_p (\mathcal{C}_{dp}(g) + \mathcal{C}_{rp}(g)) \quad (9)$$

where $\lambda_t, \lambda_p > 0$ weight transition and pushing costs, respectively; $\mathcal{C}_t$, $\mathcal{C}_{dp}$, and $\mathcal{C}_{rp}$ denote transition, direct-pushing, and recursive-pushing costs.

3) *Long-term Cost w.r.t. Goal:* Furthermore, to prioritize gaps closer to the goal, a heuristic is added to the evaluation, i.e., $h(g) = \eta \|\mathbf{o}(g) - \mathbf{x}_v^G\|$, with the scaling constant $\eta > 0$. Lastly, a short A*-style presearch is adopted to cross each candidate gap, recompute the next frontier, and continue for a limited width and depth. The *cost-to-connect* is predicted by:

$$\widehat{\text{Cost}}(g) \triangleq \mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) + \sum_{g' \in \Pi^*(g)} \{\mathcal{C}(g' \mid \cdot) + h(g')\}, \quad (10)$$

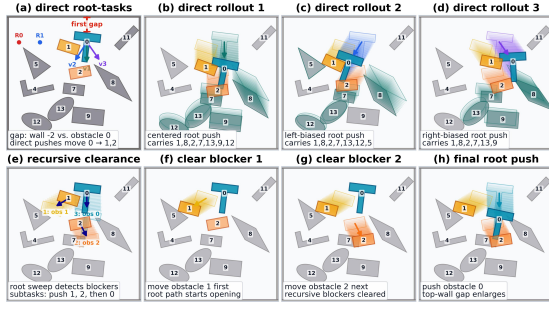


Fig. 4. **Direct** pushing moves a gap-side blocker along a gap-opening direction, whereas **recursive** pushing first clears movable blockers in the swept region and then returns to the root gap-opening task.

where $\Pi^*(g)$ is the sequence of subsequent gaps discovered after virtually crossing g . Consequently, the final output of this module is the ranked list of candidate gaps, i.e., $\text{Rank}(\Gamma_{\mathcal{L}}) \triangleq \text{argsort}_{g \in \Gamma_{\mathcal{L}}} \widehat{\text{Cost}}(g)$, in ascending predicted cost. This ordered set is passed to the hybrid search module, such that only the promising gaps are expanded first.

C. Pushing Mode Generation and Evaluation

For a selected frontier gap g , this subsection constructs a set of candidate pushing strategies $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$, where $\mathbf{v} \triangleq (v_x, v_y, \omega) \in \mathbb{R}^3$ is a short-horizon pushing direction for the movable obstacle Ω_m induced by g , and $\xi \triangleq (\mathbf{c}_m, \mathbf{u})$ is the associated pushing mode in (4). The wrench feasibility encoded in ξ acts as a screening condition rather than a force setpoint: it certifies that the chosen contact configuration can realize the required object motion within the robots' actuation limits, after which the candidate is passed to the physics rollout for validation.

1) *Direct and Recursive Pushing Tasks:* For *direct push*, let Ω_m denote one of the two movable obstacles adjacent to g . For each such obstacle, a small set of candidate body velocities $\{\mathbf{v}_m^k\}_{k=1}^{K_m}$ is sampled with a bias toward enlarging the gap g , while a few angular components are included to account for local geometry. If the swept region of a direct push intersects movable blockers, *recursive push* tasks are required instead: for each out-layer blocker Ω_m , the nominal direction moves it away from both the swept region and the reference gap sequence $\Pi^*(g)$. Recursion therefore creates a bounded dependency chain which prioritizes the final blocker Ω_m , as shown in Fig. 4.

2) *Pushing Mode Generation:* At a search state $\mathbf{s}$, a pushing mode is generated for each candidate direction $\mathbf{v}_m^k$ and obstacle Ω_m . In this subsubsection, contact tuples, object velocities, forces, and wrenches are expressed in the body frame of Ω_m . For a contact tuple $\mathbf{c}$, its multi-directional force feasibility is:

$$J_{\text{MF}}^m(\mathbf{c}, \mathbf{v}_m^k) \triangleq \sum_{\mathbf{v}_d \in \mathcal{D}_m^k} w_d \min_{\mathbf{u} \in U_m(\mathbf{c})} \|\mathbf{q}_m(\mathbf{c}, \mathbf{u}) + \mathbf{q}_m^\mu(\mathbf{v}_d)\|, \quad (11)$$

where $\|\cdot\|$ is a weighted ℓ_1 wrench norm, the finite set $\mathcal{D}_m^k$ perturbs $\mathbf{v}_m^k$, and $\mathbf{q}_m^\mu(\mathbf{v}_d)$ is the quasi-static friction wrench; the inner minimization is a small LP over the admissible forces $U_m(\mathbf{c})$. A state- and direction-dependent contact set $\mathcal{A}_m^k(\mathbf{s}) \subset \partial\Omega_m$ is drawn from the robot-reachable boundary of Ω_m by velocity-force compatibility filtering (Fig. 5), keeping a contact only if its robots reach collision-free pre-contact

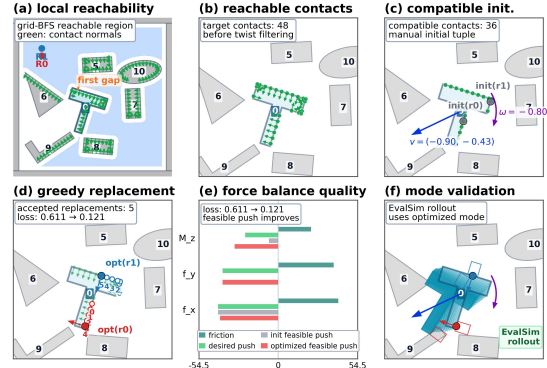


Fig. 5. Reachable contacts are filtered by the desired object twist, assigned to robots, refined by local replacement, and finally validated by simulation.

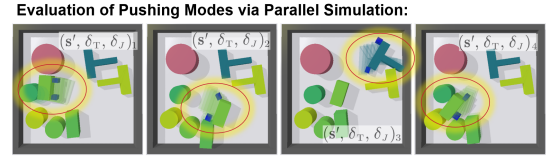


Fig. 6. Sampled push tasks are evaluated via parallel simulation, producing updated states for search expansion.

poses. Starting from cached layouts or random initializations, the tuple is refined by sequential greedy replacement: at the ℓ -th step, $\mathbf{c}_{r,c}^\ell$ replaces the r -th contact with c , and the best replacement is:

$$\mathbf{c}^* = \underset{\mathbf{c} \in \mathcal{A}_m^k(\mathbf{s})}{\text{argmin}} J_{\text{MF}}^m(\mathbf{c}_{r,c}^\ell, \mathbf{v}_m^k), \quad (12)$$

swept over contact slots until J_{MF}^m no longer decreases; a bounded backtracking variant removes the robot-order sensitivity of this replacement. If no full-team assignment exists for a task, e.g., on small faces or under the pairwise pusher-clearance constraint, one robot is released for that single push (charged a penalty when ranking modes, so full-team modes are preferred) and the full team is available again afterwards. The best tuple over initializations gives $\mathbf{c}_m^{k,*}$, from which $\mathbf{u}_m^k$ follows from the inner minimization in (11), yielding $\xi_m^k = (\mathbf{c}_m^{k,*}, \mathbf{u}_m^k)$; each mode is cached as a warm start. The best few admissible pairs are kept for gap g and passed to simulation; $\mathbf{u}_m^k$ certifies wrench feasibility while hardware tracks the contact geometry from pose/velocity commands.

3) *Pushing Strategy Evaluation:* For a current state $\mathbf{s}$ and a candidate strategy $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$, a short-horizon rollout evaluates the transition in (4) under the selected mode. As shown in Fig. 5, a strategy is accepted only if the robots reach the pre-contact poses, maintain stable contact, and induce obstacle motion above threshold. The simulation returns:

$$(\mathbf{s}', \delta_T, \delta_J) \triangleq \text{EvalSim}(\mathbf{s}, (\Omega_m, \mathbf{v}, \xi)), \quad (13)$$

where $\mathbf{s}'$ is the successor state capturing the resulting robot and object configurations; $\delta_T > 0$ is the time cost; and δ_J is the control effort. The rollout jointly simulates all robots and movable bodies, thereby capturing contact-induced object interactions. For efficiency, the pushing rollout may be bypassed when the mode has been validated before and the swept region of Ω_m is collision-free.

D. Physics-Informed Hybrid Search

The proposed physics-informed hybrid search (PIHS) selects blocking gaps and validates corresponding multi-robot pushing strategies through short-horizon simulation, as shown in Fig. 6 and summarized in Alg. 1. Unlike purely geometric NAMO planners, PIHS expands only physics-validated strategies through parallel candidate evaluation. It is a feasibility-oriented heuristic search, where Eq. (6) defines the ranking objective for candidate schedules. **The search operates on a finite sampled candidate set and therefore does not guarantee global optimality or completeness of the underlying continuous hybrid planning problem.**

1) *Tree and Initialization:* The search tree $\mathcal{T}$ holds nodes $\nu \triangleq (\mathbf{s}, \pi)$, where $\mathbf{s}$ is the system state of all robots and movables in (4) and π the partial pushing strategy in (6). Two global functions are maintained: $\text{Rank}(\nu)$ stores the ranked candidate gaps $\Gamma_{\mathcal{L}_\nu}$ of the frontier loop $\mathcal{L}_\nu$ together with their exploration costs by (10), and $\chi(\nu)$ returns the cumulative execution cost from the root to ν . The root is $\nu_0 \triangleq (\mathbf{s}_0, \emptyset)$ with initial state $\mathbf{s}_0$.

2) *Node Selection:* At each iteration, the node with minimum best-first priority is selected from the queue. Priority balances realized cost and estimated remaining effort:

$$f(\nu) \triangleq \chi(\nu) + \min_{g \in \text{Rank}(\nu)} \widehat{\text{Cost}}(g), \quad (14)$$

where $\chi(\nu)$ is the cumulative execution cost so far, and the second term is the minimum predicted remaining cost among the unexplored gaps of ν . **The remaining-cost term is a heuristic and is not required to be admissible. The score prioritizes nodes by realized cost and promising future actions, but it does not provide an A* optimality guarantee.**

3) *Node Expansion with Parallel Simulation:* Let the selected node be $\nu \triangleq (\mathbf{s}, \pi)$. For each ranked gap $g \in \text{Rank}(\nu)$, its predicted gap sequence $\Pi^*(g)$ in (10) is first derived. Then, the mode optimization by (11) provides a candidate strategy set $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$, including both the direct push of the adjacent movable obstacles, and recursive push of out-layer obstacles. Thus, expansion is prioritized first by the gap ordering in $\text{Rank}(\nu)$, and then by the candidate ordering within Ξ_g . Each candidate strategy $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$ is first checked by a geometric quick-pass and, if it passes, evaluated by the short-horizon rollout in (13). A successful evaluation yields $(\mathbf{s}', \delta_\tau, \delta_\mathcal{J})$, from which a child node $\nu' \triangleq (\mathbf{s}', \pi')$ is created by appending $\tau \triangleq (\Omega_m, \mathbf{v}, \xi)$ to the current partial schedule π . The cumulative cost is updated by: $\chi(\nu') \triangleq \chi(\nu) + \widehat{\text{C}}(\nu, \tau)$, where $\widehat{\text{C}}(\nu, \tau) \triangleq \delta_\tau + \alpha \delta_\mathcal{J}$ is the realized incremental cost. Note that all candidate strategies in the current batch are evaluated *concurrently in parallel*, and a single expansion can produce multiple validated successors.

4) *Termination:* Termination occurs when a node ν reaches a state $\mathbf{s}$ for which a W -clear path $\mathcal{P}_V^W$ exists from $\mathbf{x}_V^S$ to $\mathbf{x}_V^G$ by (8). **Thus, the schedule π stored in ν is a feasible solution satisfying the terminal clearance condition and the validated pushing transitions. It encodes both the sequence of gaps to clear and the pushing strategies that realize them, without implying that $\mathcal{J}_{\text{ch}}(\pi)$ is globally minimal. The objective covers the clearing phase only, since the external vehicle**

Algorithm 1: Physics-Informed Hybrid Search

Input: $\mathbf{s}_0, \alpha, \text{EvalSim}(\cdot), W$ -criterion by (8)
Output: π^*

```

1  $\nu_0 \leftarrow (\mathbf{s}_0, \emptyset)$ ,  $\chi(\nu_0) = 0$ ;
2 Init  $\mathcal{Q}$  by (14);
3 while not terminated do
4    $\nu \leftarrow \mathcal{Q}.\text{pop\_min}()$  by (14);
5   foreach  $g \in \text{Rank}(\nu)$  do
6     Compute  $\Pi^*(g)$ , generate  $\Xi_g$ ;
7     foreach  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$  do
8        $\tau = (\Omega_m, \mathbf{v}, \xi)$  // Direct or recur.
9        $(\mathbf{s}', \delta_\tau, \delta_\mathcal{J}) \leftarrow \text{EvalSim}(\nu, \tau)$  by (13);
10      if success then
11         $\nu' \leftarrow (\mathbf{s}', \pi \cup \{\tau\})$ ;
12         $\chi(\nu') \leftarrow \chi(\nu) + \delta_\tau + \alpha \delta_\mathcal{J}$ ;
13        Add  $\nu'$  to  $\mathcal{Q}$ ;
14      if  $\mathbf{s}'$  admits  $\mathcal{P}_V^W$  by (8) then
15         $\pi^* \leftarrow \pi'$ , break;
16 return  $\pi^*$ ;

```

traverses the corridor after clearing completes; traversal time is therefore not part of T .

Remark 2. Efficiency comes from three lightweight heuristics: gap-lookahead reduces unproductive branches, MODE-CACHE warm-starts mode optimization, and the quick-pass rollout bypass skips simple motions whose mode was previously validated, avoiding redundant simulations. ■

E. Overall Analyses

1) *Execution and Online Adaptation:* The hybrid search returns a pushing schedule $\pi = \{\tau_1, \dots, \tau_K\}$, with $\tau_k \triangleq (\Omega_k, \mathbf{v}_k, \xi_k)$. The schedule is executed in a receding-horizon manner: for each τ_k the robots reach the mode ξ_k and follow the pushing direction $\mathbf{v}_k$ from the current state, as in (13). **On the hardware the optimized contact forces certify feasibility rather than being commanded: a mode is retained only if its object wrench lies within the robots' actuation limits, and the controller tracks the planned contact poses and pushing velocities with feedback. Planning therefore runs in simulation through EvalSim, and every executed push returns an observed state that re-enters the planner, so that physical execution continuously feeds back into the search.** During transitions, robots are reassigned by preserving the circular order of contact points along the obstacle boundary [7], reducing path crossings under sufficient clearance. After each push the state is updated from observation, the WCCG is rebuilt from the current configuration, and execution continues until a W -clear path $\mathcal{P}_V^W$ connects $\mathbf{x}_V^S$ and $\mathbf{x}_V^G$. Replanning is triggered by observed execution discrepancies – failed contact establishment, stalled pushing, or invalidated WCCG connectivity – in which case Alg. 1 is called from the current state to derive the updated schedule.

2) *Computational Complexity:* Given M movable obstacles, the computational cost of one node expansion in Alg. 1 is dominated by candidate generation and physics validation. The WCCG update and gap ranking incur $\mathcal{O}(M \log M + |\mathcal{V}| + |\mathcal{E}_c| + |\mathcal{E}_b|)$ and $\mathcal{O}(|\Gamma_{\mathcal{L}_\nu}| \log |\Gamma_{\mathcal{L}_\nu}|)$ costs, respectively, while the dominant **The dominant simulation cost**

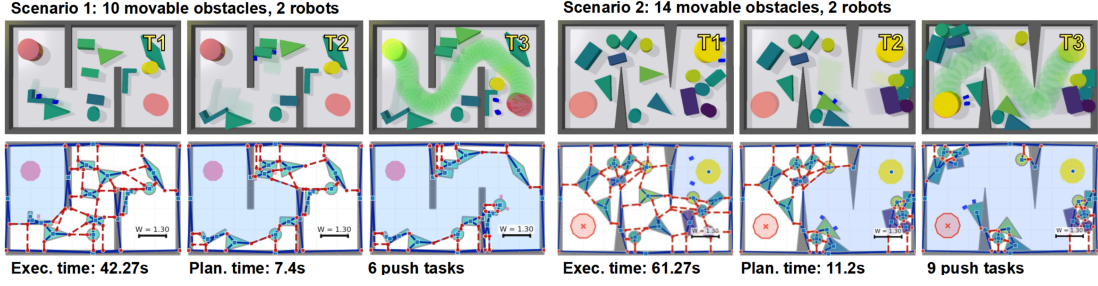


Fig. 7. Execution and planning alignment in Scenario 1 (Left) and Scenario 2 (Right). **Top:** simulated snapshots during path clearing. **Bottom:** corresponding WCCG overlays with the start face in blue. The executed pushes progressively expand the reachable W -clearance region until the goal is connected.

TABLE I
WCCG CONNECTIVITY VALIDATION AND MEAN RUNTIME.

Type	CC=FR	FP/FN [†]	Time (ms)		
			CC	FR	WCCG
Convex	100/100	0/0	1.23	282.9	0.86
Non-convex	98/100	0/0	1.51	204.1	1.30
Mixed	98/100	0/0	1.33	242.4	1.05
Total	296/300	0/0	1.36	243.1	1.07

[†]WCCG errors on CC-FR-consistent cases. All runtimes are measured in one harness over the same 300 scenes, with WCCG served by its compiled backend;

$\mathcal{O}(\sum_{g \in \text{Rank}(\nu)} |\Xi_g| (C_{LP} + \rho_\nu T_{\text{sim}}/n_w))$ grows with the number of validated candidates and is reduced by the n_w parallel workers, so that scalability is mainly determined by validated pushing candidates, while quick-pass filtering and parallel simulation reduce the practical expansion cost. Robot-team size mainly affects mode generation, where greedy contact refinement avoids exhaustive $|A|^N$ contact assignment and evaluates bounded single-contact replacements.

IV. EXPERIMENTS

Numerical experiments are conducted to validate the proposed PIHS algorithm. The algorithms are implemented in Python3 and PyBullet [22] on a computer with an Intel Core i9-14900KF and an NVIDIA RTX 4080. Videos are provided as multimedia attachments.

A. Numerical Simulations

1) *Setup and Configuration:* The simulation runs at a step size of $\Delta t = 1/240$ s with a control period of $1/40$ s; robots are disk or box pushers consistent with the W -clearance definition, and movable obstacles have masses sampled from $[5, 15]$ kg. A trial succeeds when the external vehicle reaches its goal through a W -clear path. The nominal scenario is an 8×5 m workspace with bottleneck-forming bar obstacles, where two robots clear 10–15 movable obstacles under minimum-separation constraints. Unless stated otherwise, comparisons and ablations use Scenario 1 (10 movable obstacles, 40 trials), and Scenario 2 (14 obstacles) provides an additional cluttered case. All methods share the robot footprints, obstacle configurations, collision and reachability checks, and execution simulator, with the same per-trial timeout of 800 s and planning budget of 20,000 control steps; unsuccessful trials are assigned the 800 s cap in the planning-time statistics.

2) *Results:* Two representative simulation runs are shown in Fig. 7. In Scenario 1 the start and goal lie in different W -clearance faces, so the planner clears the blocked passage with 6 pushes in 7.4 s and execution finishes in 42.3 s. Scenario 2 requires a 9-push plan in 11.2 s and 61.3 s of execution. In both

TABLE II
PERFORMANCE COMPARISON AND ABLATIONS ON SCENARIO 1 OVER 40 RANDOMIZED TRIALS.

Method / Variant	Succ.(%)	PT (s)	ET (s)	#Sims	#Pushes
DFS-WCCG	25.0	>100.0	51.2	>500	8.0
SL-Push (offline)	62.5	0.06	44.1	0.0	7.3
SL-Push (sim)	75.0	18.2	64.6	20.0	10.2
Rec-NAMO	37.5	13.3	42.4	0.0	7.8
PushAround (ours)	97.5	10.3	28.6	121.5	6.0
w/o gap-lookahead	82.5	16.9	35.2	178.3	7.2
w/o mode-cache	92.5	17.1	33.4	164.6	6.5
w/o quick-pass	90.0	23.6	41.9	208.2	6.1
w/o reinsertion	85.0	12.8	31.1	151.7	7.8



Fig. 8. Failure cases by baselines Rec-NAMO, SL-Push and DFS-WCCG.

cases the executed pushes progressively enlarge the reachable W -clearance region until a complete W -clear path is found, as shown by the expanding blue start face in the WCCG overlays.

3) *WCCG Connectivity Validation:* WCCG is compared against two references on 300 random scenes covering convex, non-convex and mixed obstacles (Table I). The continuous reference (CC) follows the classical configuration-space construction, inflating obstacle and workspace boundaries by $W/2$ and querying connectivity with a fine continuous collision check at 0.01 m oracle resolution instead of a fixed grid [23]; the rasterized reference (FR) discretizes the same inflated workspace at 0.02 m for an occupancy-grid query. CC and FR disagree in 4 of the 300 cases, which bounds the ambiguity of the ground truth itself; on the 296 scenes where they agree, WCCG matches CC with zero false positives and zero false negatives. A query costs 1.07 ms against 1.36 ms (CC) and 243 ms (FR), while additionally supplying the gap graph needed by PIHS.

4) *Comparisons:* Four baselines run on Scenario 1 over 40 randomized trials under the same W -clearance criterion, object distribution and simulator: **DFS-WCCG** (WCCG-guided search with fixed pushes), **SL-Push** (offline and simulation-validated variants), and **Rec-NAMO** (recursive NAMO) [3]–[5], [16]. Table II shows that PushAround reaches the highest

TABLE III
SCALABILITY AND ROBUSTNESS UNDER OBSTACLE DENSITY,
ROBOT-TEAM SIZE, AND EXECUTION-SIDE PHYSICS MISMATCH.

Case	Trials	Succ.(%)	PT (s)	ET (s)
<i>Obstacle density</i> (PushAround, $N=2$, nominal)				
$M = 15$	10	100	2.5 ± 0.9	37.1 ± 21.0
$M = 30$	10	100	11.7 ± 3.7	100.9 ± 34.7
$M = 45$	10	100	53.4 ± 24.5	272.7 ± 97.4
<i>Robot-team size</i> ($M=30$, nominal; 10 trials per cell)				
PushAround, $N=3$	10	100	30.0 ± 12.7	97.3 ± 42.1
PushAround, $N=4$	10	100	29.2 ± 8.9	110.1 ± 33.1
SL-Push, $N=3$	10	80	30.0 ± 46.3	193.3 ± 111.3
SL-Push, $N=4$	10	80	23.4 ± 18.2	171.7 ± 61.2
DFS-WCCG, $N=3$	10	100	142.6 ± 183.1	178.1 ± 72.1
DFS-WCCG, $N=4$	10	90	101.9 ± 103.9	175.9 ± 113.6
<i>Physics mismatch</i> ($M=30$, $N=2$)				
Nominal	10	100	16.8 ± 3.4	104.7 ± 37.4
Mass $\times 0.6-1.4$	40	100	16.8 ± 5.4	105.9 ± 44.6
Lateral fric. $\times 0.6-1.4$	40	97.5	16.5 ± 5.7	114.0 ± 48.5
Spin fric. $\times 0.6-1.4$	40	97.5	15.2 ± 3.6	100.1 ± 35.7
Force limit $\times 0.6-1.4$	40	95.0	18.9 ± 10.9	115.9 ± 58.5
All perturbed	160	97.5	16.9 ± 7.0	108.9 ± 47.4

Each perturbed row scales one parameter; the others stay nominal.

success rate (97.5%) and the lowest combined planning-and-execution time (38.9 s). DFS-WCCG needs more than 100 s of planning (exponential branching); SL-Push (offline) plans in 0.06 s but many displacements fail in execution (62.5%); SL-Push (sim) reaches 75.0% at 18.2 s planning; and Rec-NAMO usually fails to build a connected W -clear corridor in dense clutter, as illustrated in Fig. 8.

5) *Ablation Studies*: Four internal ablations are conducted: **w/o gap-lookahead** (gaps ranked without downstream costs), **w/o mode-cache** (no warm start from validated modes), **w/o quick-pass** (all candidates evaluated by rollout), and **w/o reinsertion** (failed nodes not reconsidered). Table II shows that gap-lookahead and reinsertion mainly improve robustness (success drops by more than 10% when either is removed), whereas mode-cache only lowers success to 92.5%; quick-pass affects efficiency, with simulations rising from 121.5 to 208.2 and planning from 10.3 to 23.6 s.

6) *Scalability and Robustness Analysis*: Generalization is further evaluated in the same 8×8 m workspace by scaling the obstacle density M , the robot-team size N , and the execution-side physics parameters (Table III and Fig. 10). (I) **Obstacle density**: with two robots and 10 trials per density, PushAround keeps a 100% success rate for all $M \in [15, 45]$ while its mean planning time grows from 2.5 to 53.4 s, whereas all baselines solve none of the $M=45$ trials and removing recursive pushes lowers success to 70%. (II) **Robot-team size**: at $M=30$, PushAround solves all ten trials at $N=2, 3$ and 4 after two mode-generation refinements: bounded backtracking contact assignment removes the robot-order sensitivity of greedy replacement, and a temporarily released robot is re-activated once a full-team assignment exists, with a penalty favoring full-team modes. The fallback belongs to the shared mode generator, so the baselines benefit from it as well, yet at $N=4$ PushAround stays the most economical with 29.2 s planning, 110.1 s execution and 10 pushes versus 101.9 s, 175.9 s and 14 for DFS-WCCG. (III) **Physics mismatch**: the planner keeps the nominal model and only the execution simulator is perturbed, one parameter at a time: PushAround succeeds in 156 of 160 perturbed trials (97.5%), the four failures being

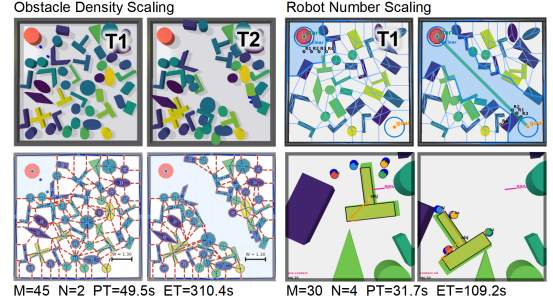


Fig. 9. Scalability of the clearing process. **Left**: obstacle-density scaling ($M=45$ objects, two robots): snapshots during the clearing (**top**) and WCCG overlays with the start face in blue (**bottom**). **Right**: robot-team scaling ($M=30$ objects, four robots): initial and final states (**top**) and zoomed views of one push before contact and during execution (**bottom**).

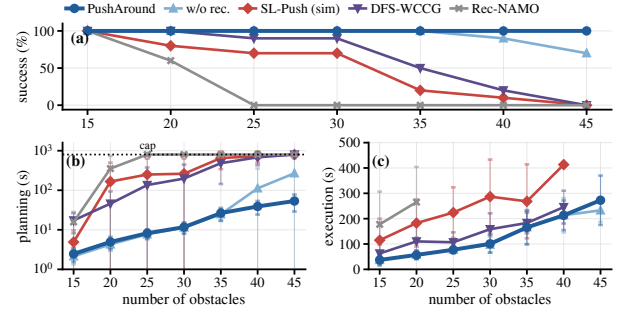


Fig. 10. Obstacle-density scaling (8×8 m, 10 independent seeds per density). (a) Success rate; (b) mean planning time, with unsuccessful trials capped at the 800 s budget (dotted line); (c) mean execution time of the successful trials (Rec-NAMO has none beyond $M=20$). Error bars denote one standard deviation across seeds.

one low-friction planning failure and three budget timeouts. Under the same $\pm 40\%$ perturbations SL-Push (sim) averages 32.5% and DFS-WCCG 57.5%, at an order of magnitude higher planning time.

B. Hardware Experiments

1) *System Description*: Experiments use a 5×6 m workspace of interlocking 0.6×0.6 m foam mats (Fig. 11), 2 UGVs and 6 cardboard-box movable obstacles. Robots and movables each carry three to four motion-capture markers whose global poses are streamed to PyBullet in real time. The two Mecanum UGVs are velocity-controlled, and the optimized contact forces are used only for feasibility evaluation rather than direct motor control. Motion capture provides the current robot and object poses, which are compared with the successor state predicted by EvalSim: a push triggers replanning only when the measured object position error exceeds 0.3 m or its orientation error exceeds $\pi/4$, a deliberately loose threshold that tolerates slight over-push or off-axis push. Movables are randomly placed for each trial.

2) *Results*: Two representative real-world runs are shown in Fig. 11. In Scenario 1, PIHS breaks five candidate gaps within 6.5 s using 12 node expansions, 49 visited snapshots and search depth 6; the four-task schedule first rotates the yellow L-shaped object to open a gap near an immovable obstacle and then separates the green and orange objects until a W -clear path is established at $t = 52$ s. In Scenario 2, PushAround succeeds in 8/10 trials with an average of 5.2 pushes and 11.4 s of planning; it first selects a direct push on the blue

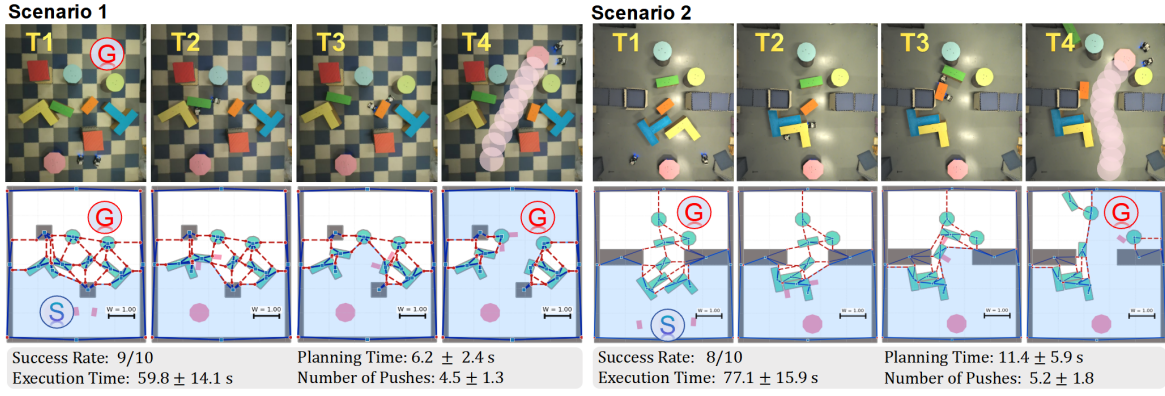


Fig. 11. Real-world pushing experiments with execution-planning alignment (over 20 runs and 2 scenarios). **Top:** snapshots of two robots pushing obstacles in a 5×6 m workspace with movable objects and fixed boundaries. **Bottom:** WCCG overlays with the start face in blue.

L-shaped object, which also moves the adjacent T-shaped block and opens the first gap into the narrow passage. This neighboring-object motion is not a separately encoded action but an emergent outcome of the coupled rigid-body rollout in EvalSim, whose snapshot is used to update the WCCG and continue replanning. Execution takes 77.1 s on average, with 15.9 s of deviations from direction-dependent force limits, drifting and Mecanum-wheel slippage, all compensated by online replanning.

V. CONCLUSION

This work presents PushAround, a multi-robot framework for path clearing in unstructured environments that jointly plans obstacle selection, contact points and pushing forces through a hybrid search, and demonstrates hardware feasibility with state-feedback replanning. Future directions include obstacle-state estimation without external sensing, uncertain masses, and limited robot visibility.

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